

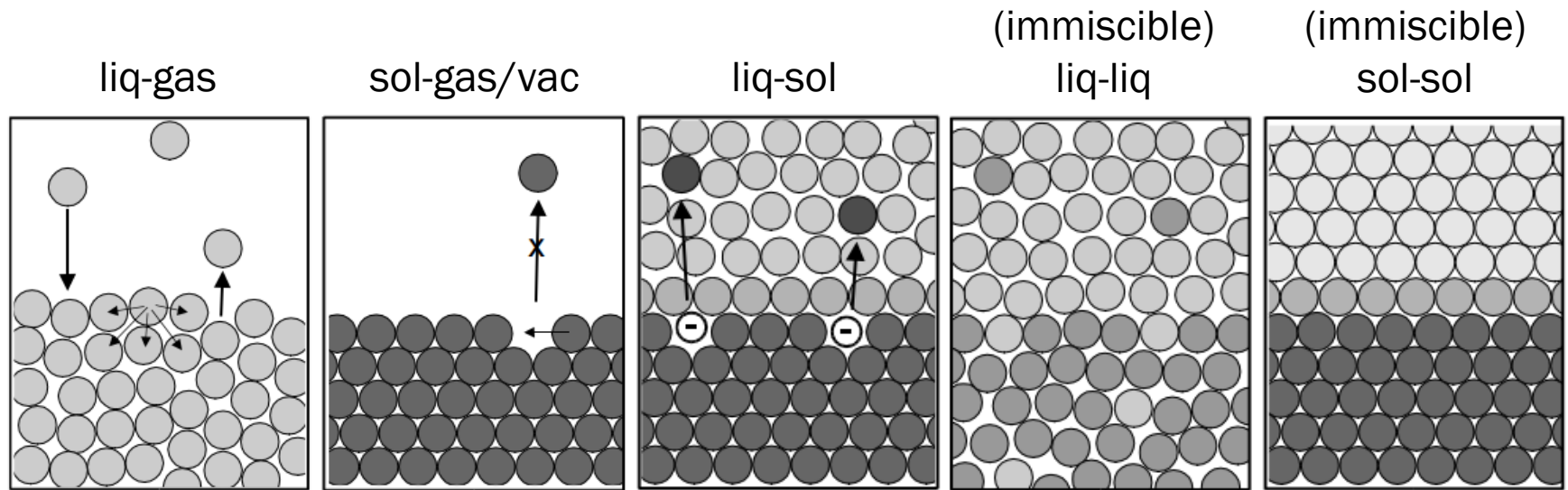
Physical Chemistry of Materials

Lecture 16

Applications of Solid-State Chemistry

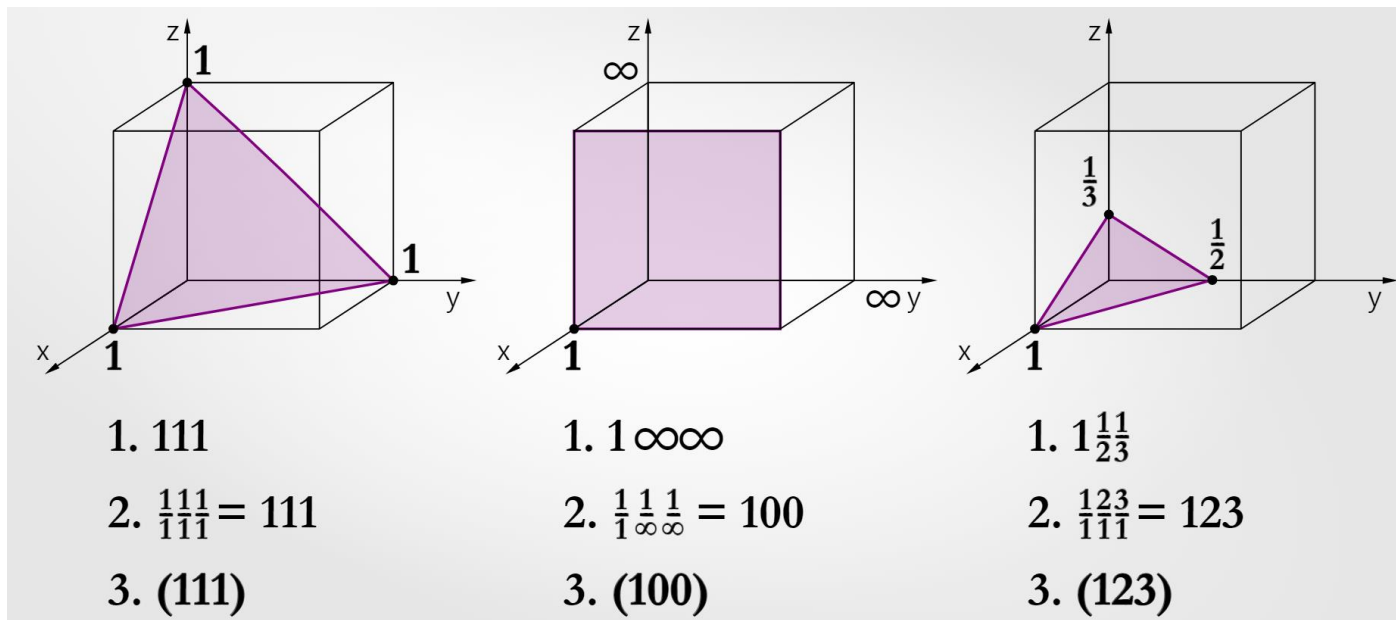


Surfaces and Interfaces



Miller Indices

Reciprocals of the intersection distances along lattice axes

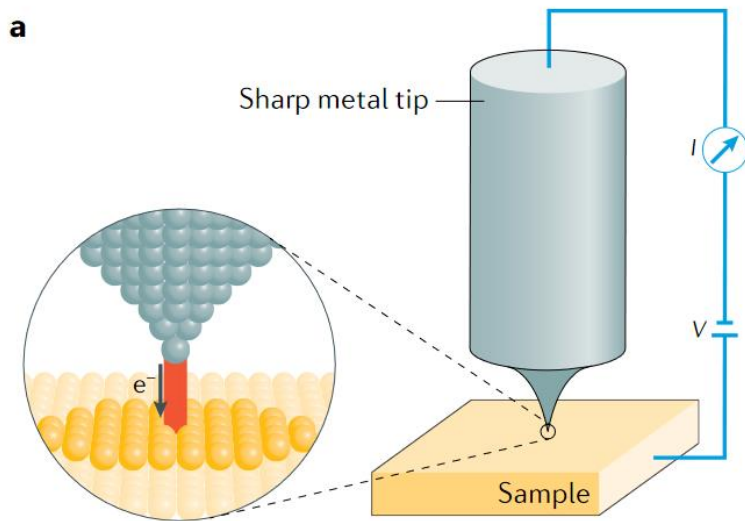


Integers!

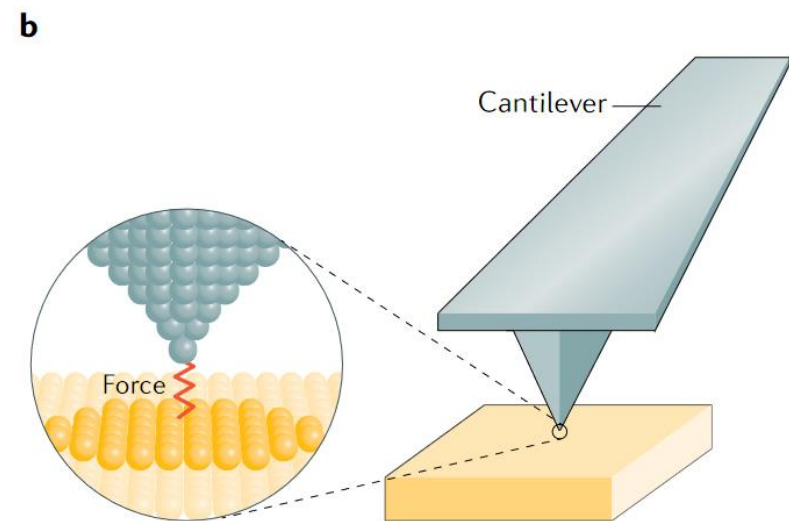
(Image: <https://mstudent.com/miller-indices/>)

Last Class

Characterization: Microscopy



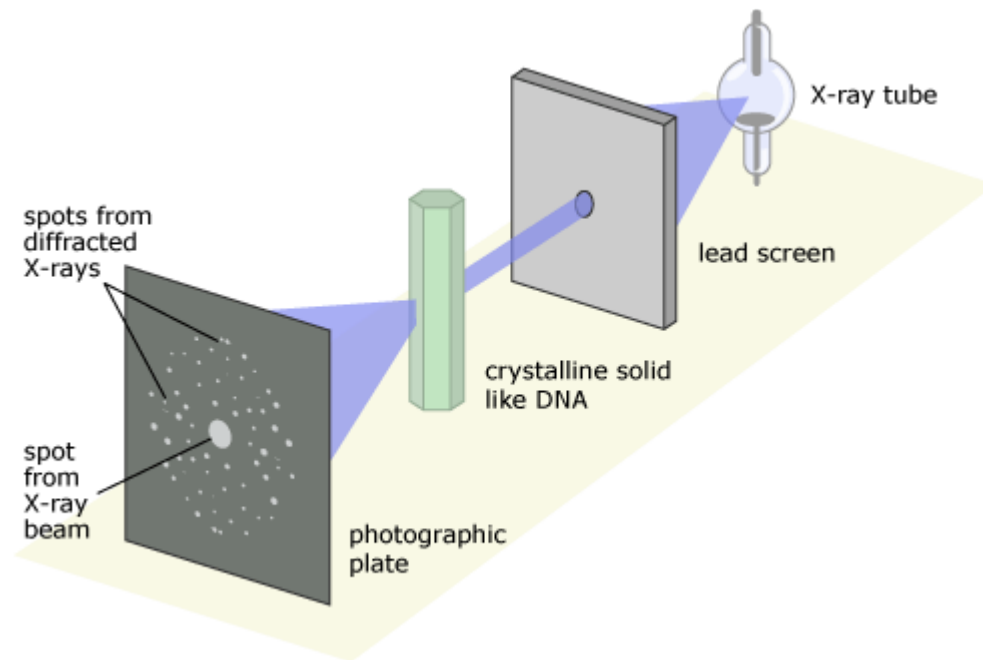
Surface Tunneling Microscope (STM)



Atomic Force Microscope (AFM)

Last Class

Characterization: X-ray Diffraction

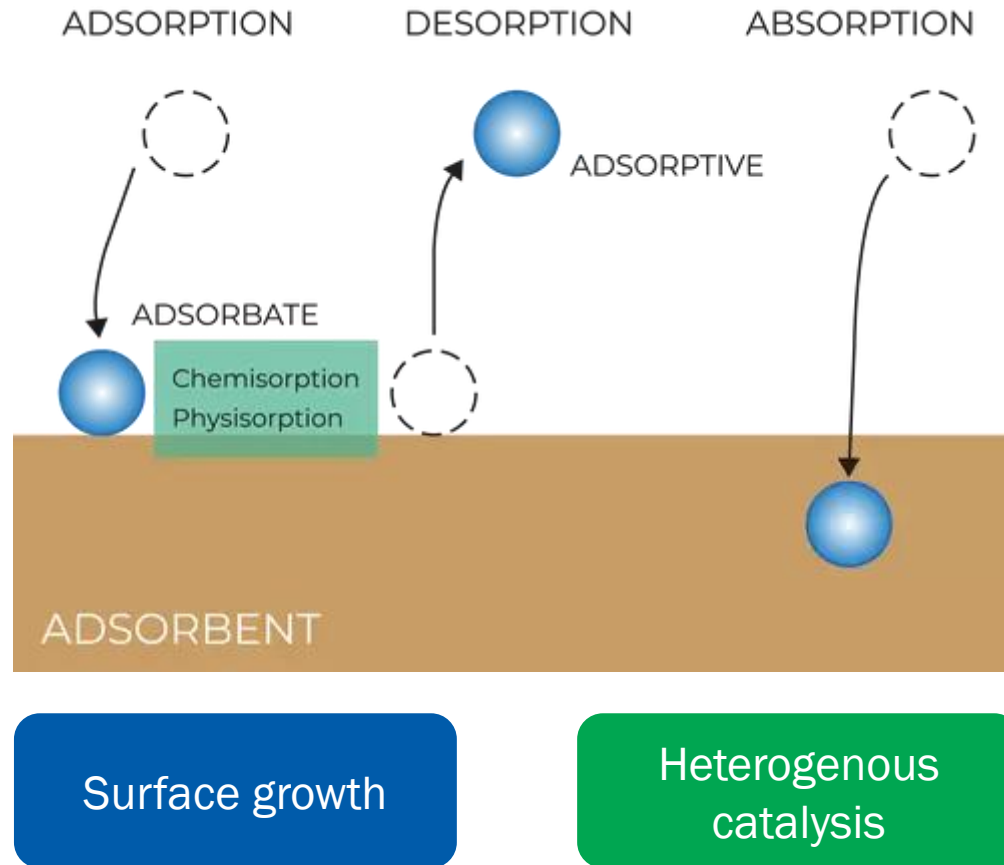


Real Crystal Surfaces

- Rearrangement
 - Relaxation
 - Reconstruction
- Defects
 - Point and line defects
 - Dislocations
 - Grain boundaries

Last Class

Adsorption and Surface Reaction



(Image: <https://www.microtrac.com/applications/knowledge-base/what-is-adsorption/>)

Fractional Coverage

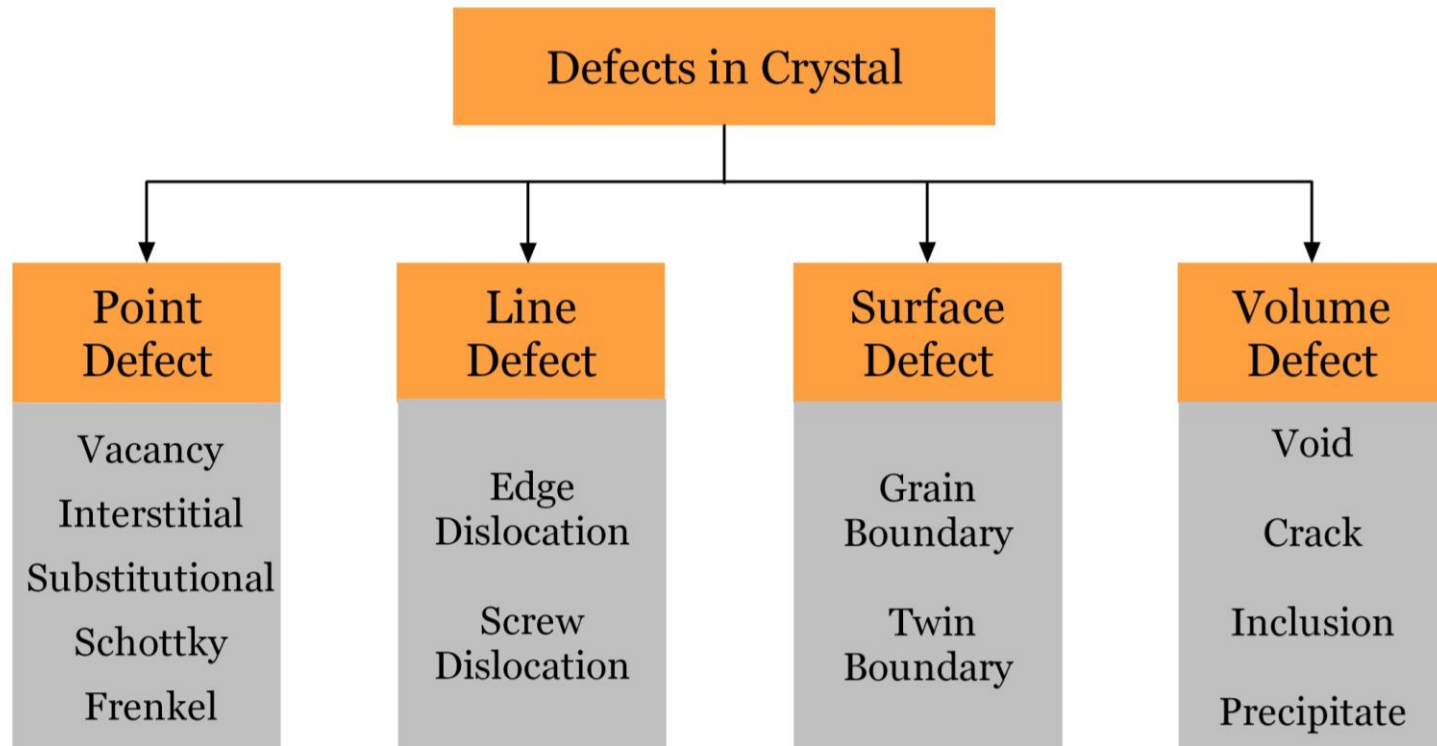
$$\theta = \frac{\text{number of adsorption sites occupied}}{\text{number of adsorption sites available}}$$

can be used to study [adsorption kinetics](#)
instead of partial pressure and concentration

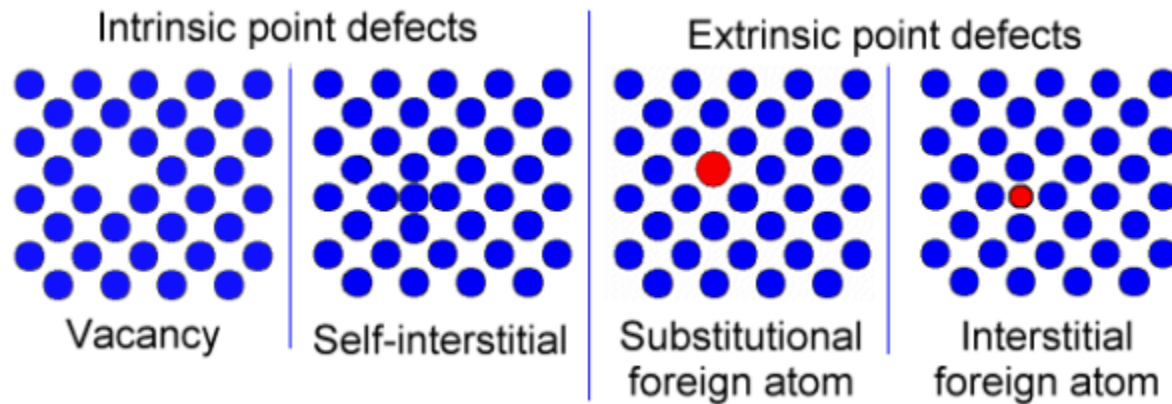
Today's Topics

1. Defects and doping: pp. 10-14
2. Band bending: pp. 15-23
3. Nanomaterials: pp. 24-34

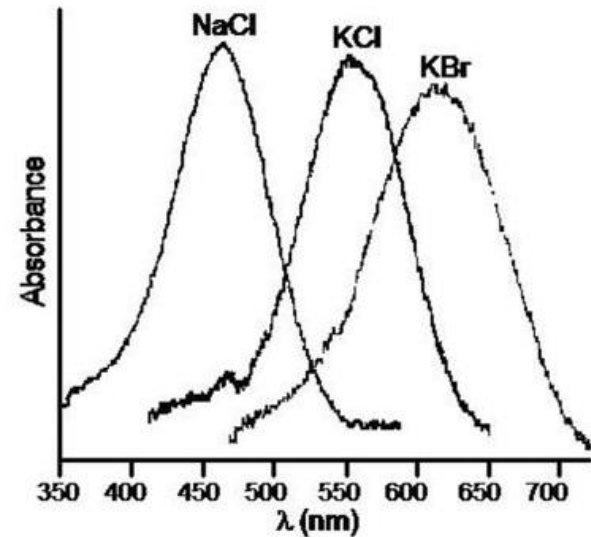
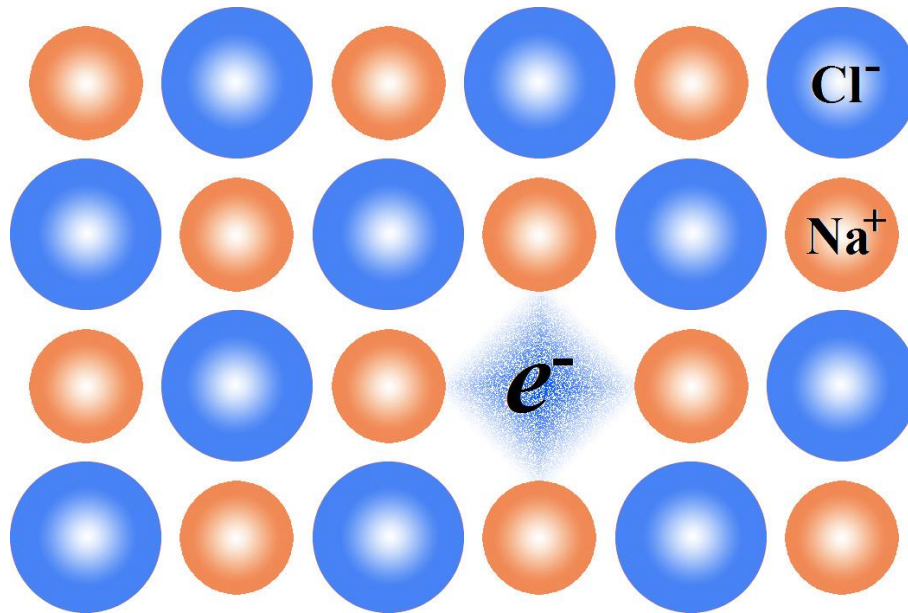
Defects



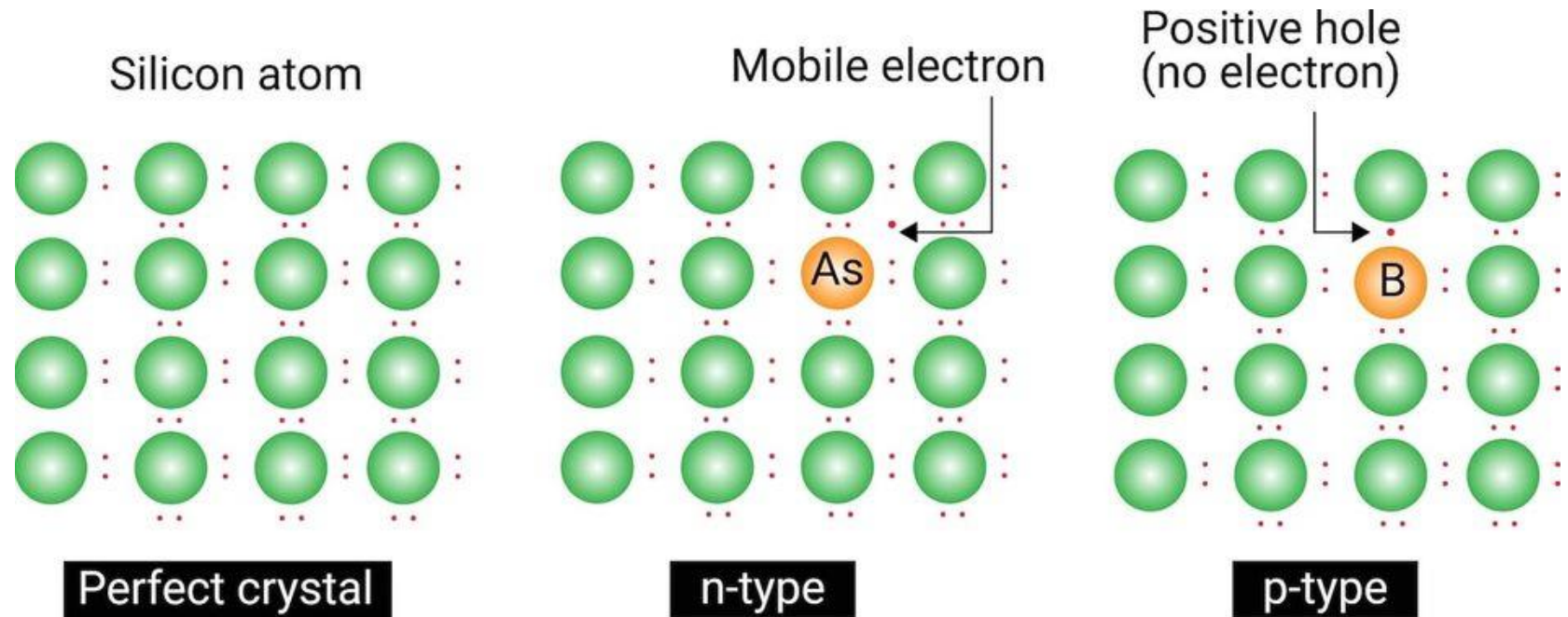
Point Defects



F Centers

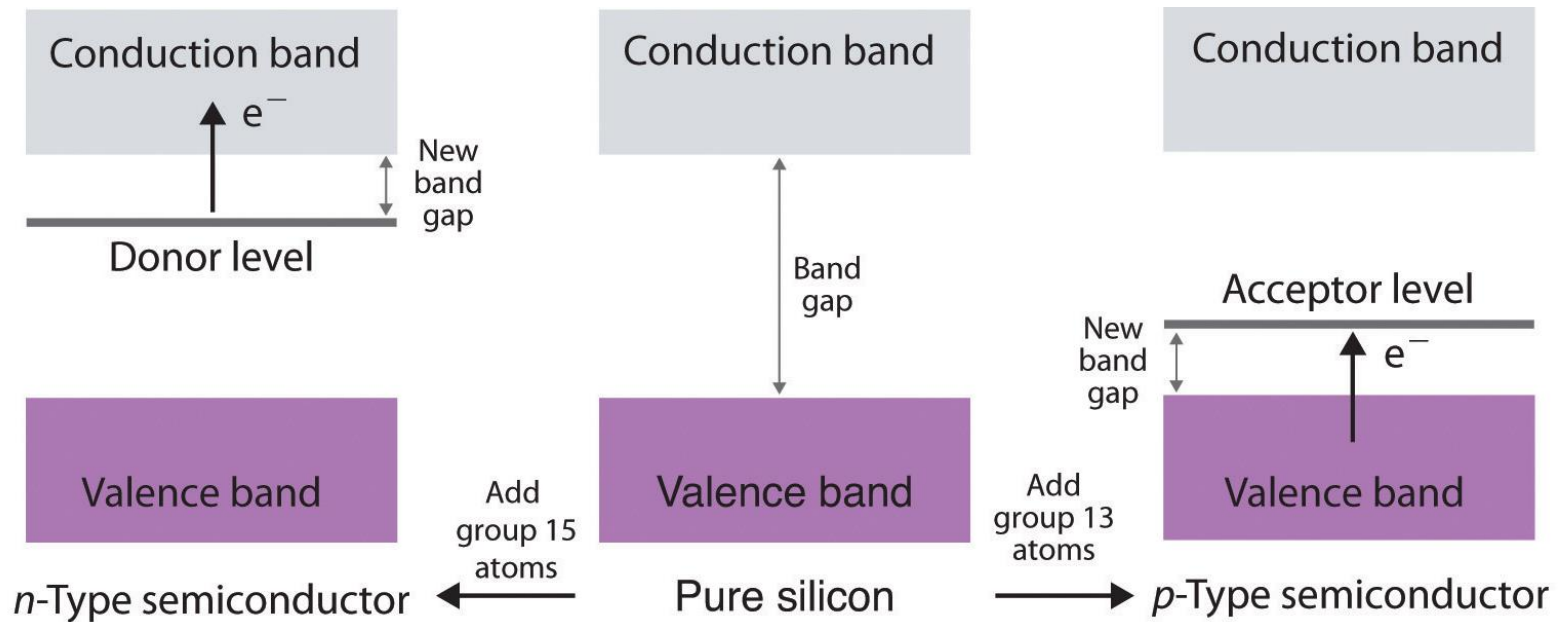


Defects and Semiconductors

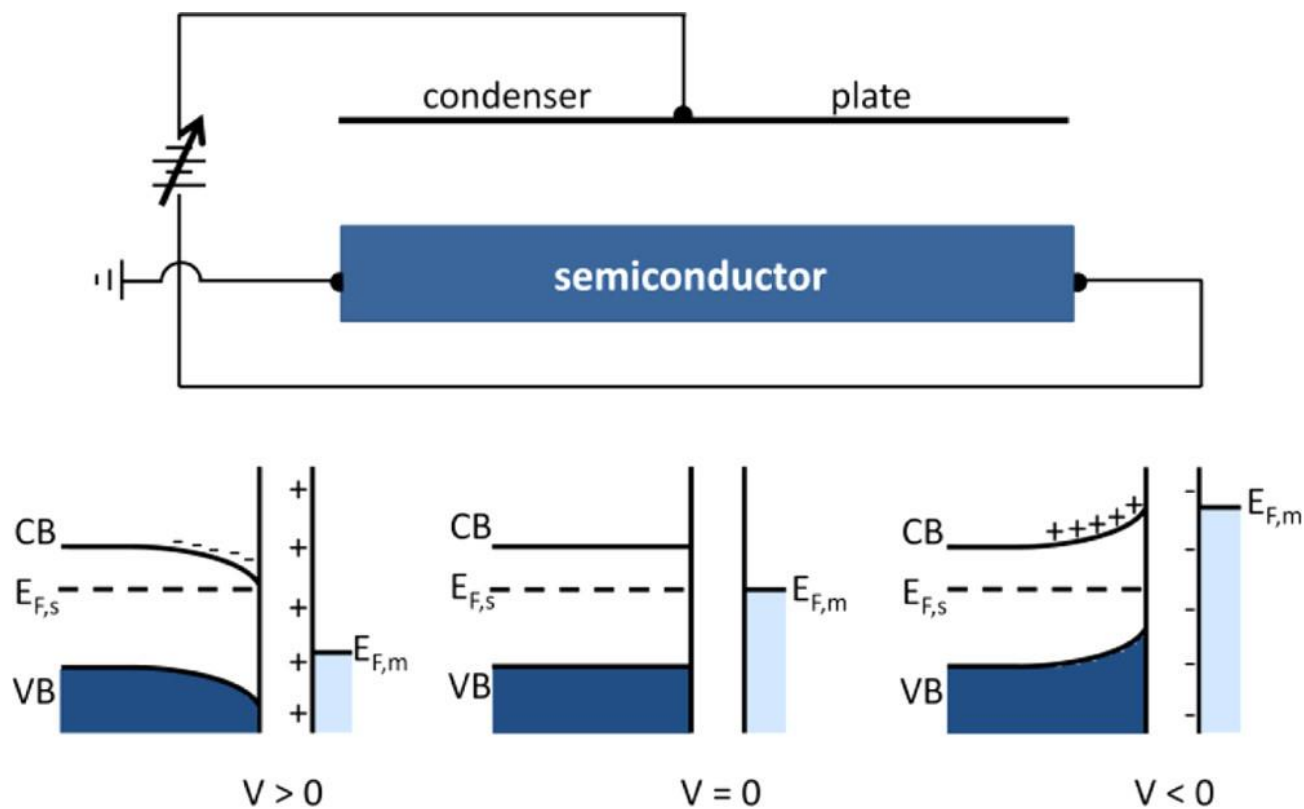


(Image: <https://www.wevolver.com/article/understanding-n-type-vs-p-type-semiconductors>)

Defects and Semiconductors

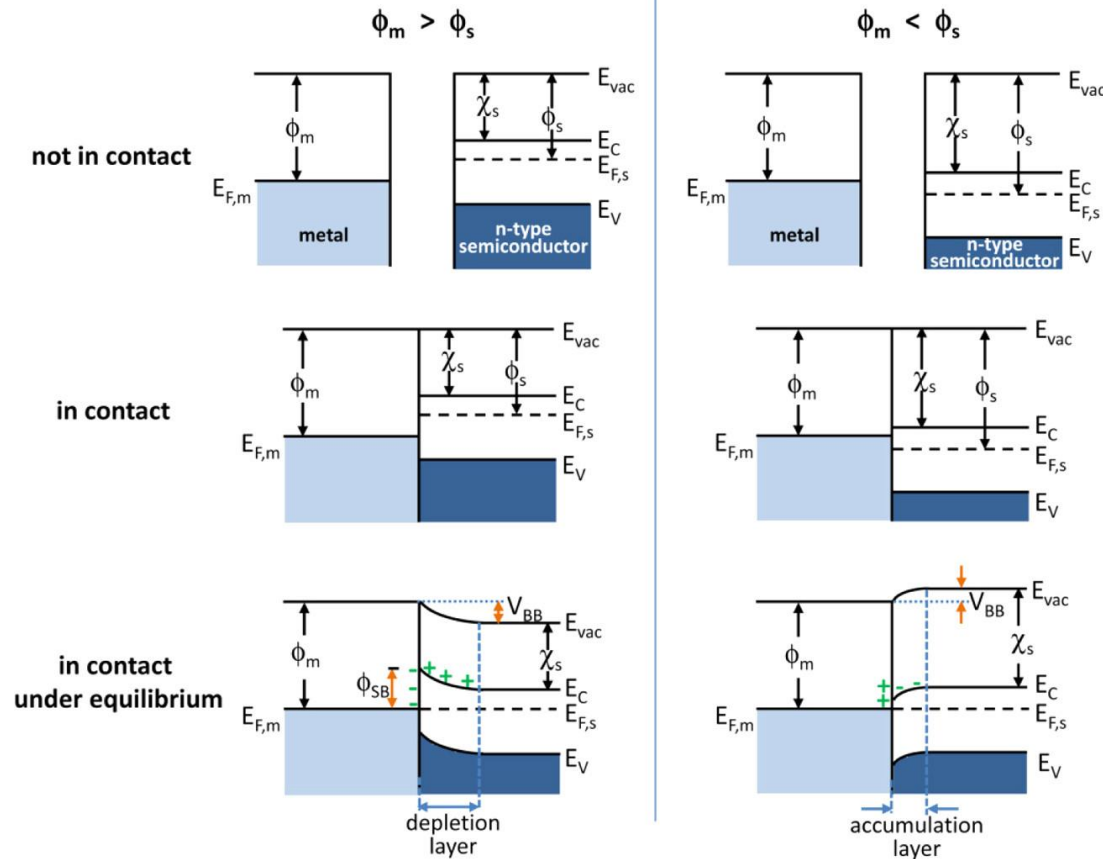


Band Bending



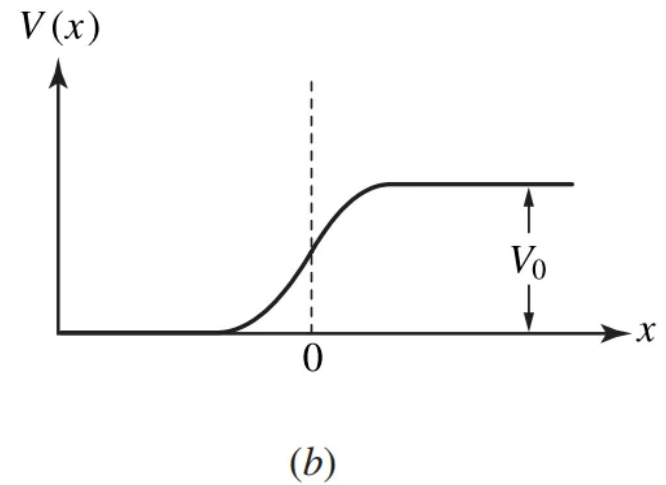
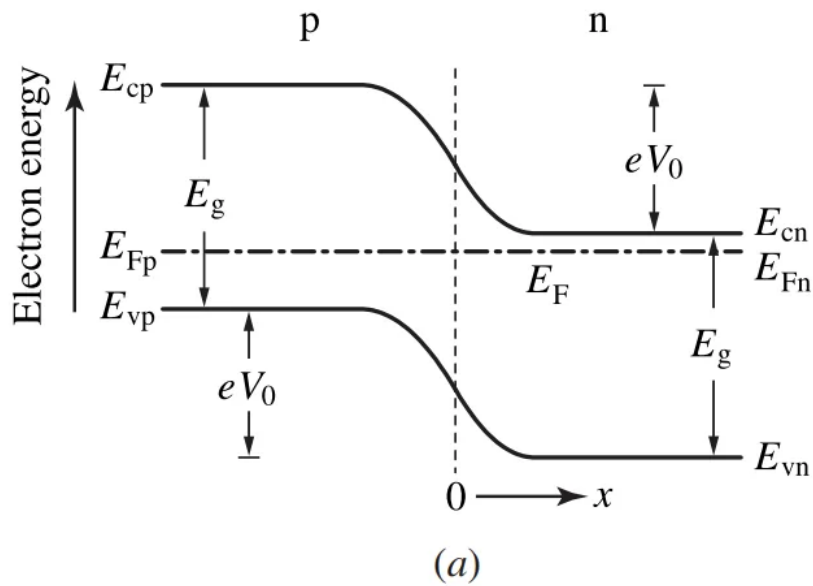
(Image: Zhang and Yates, *Chem Rev* (2012))

Band Bending at Interface

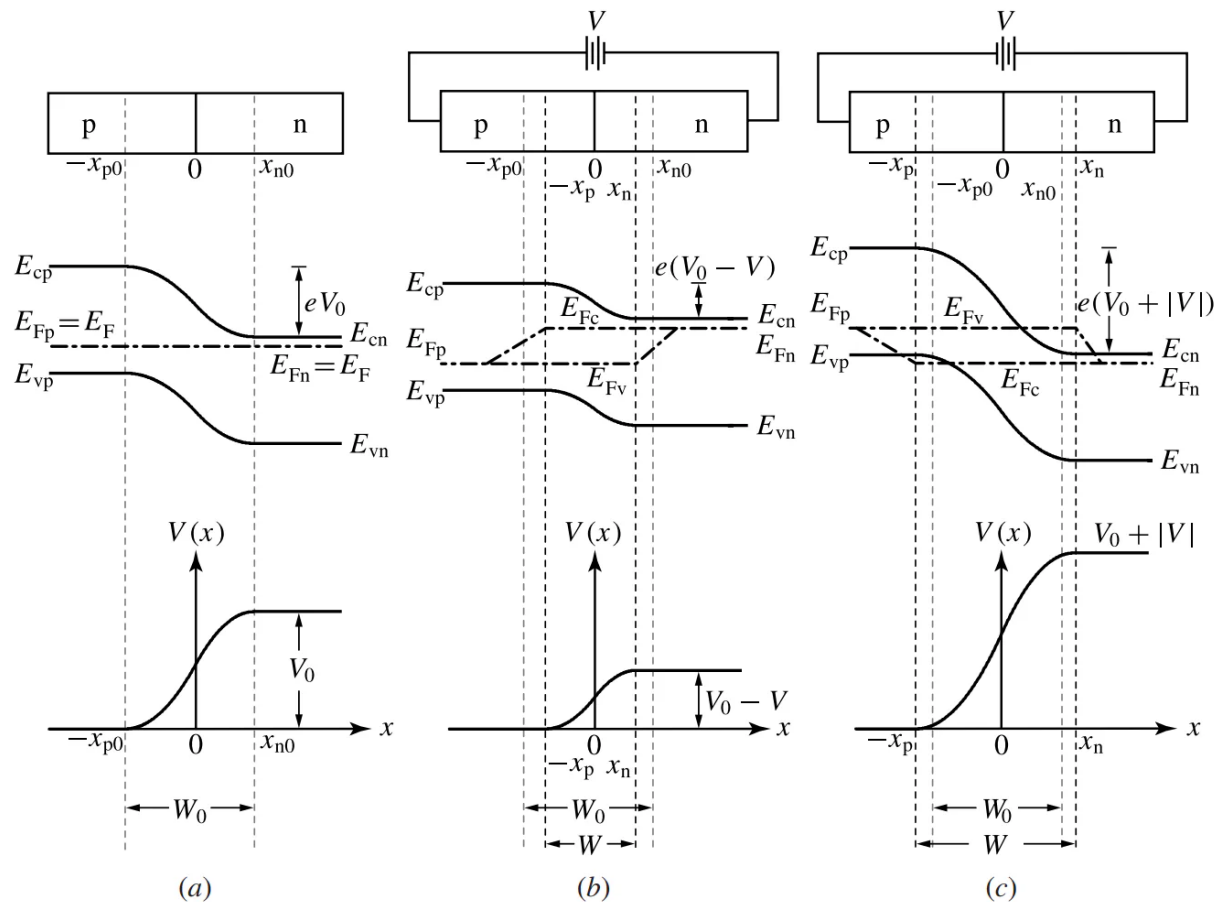


(Image: Zhang and Yates, *Chem Rev* (2012))

p-n Junction

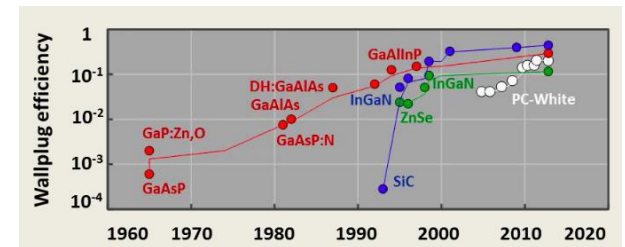
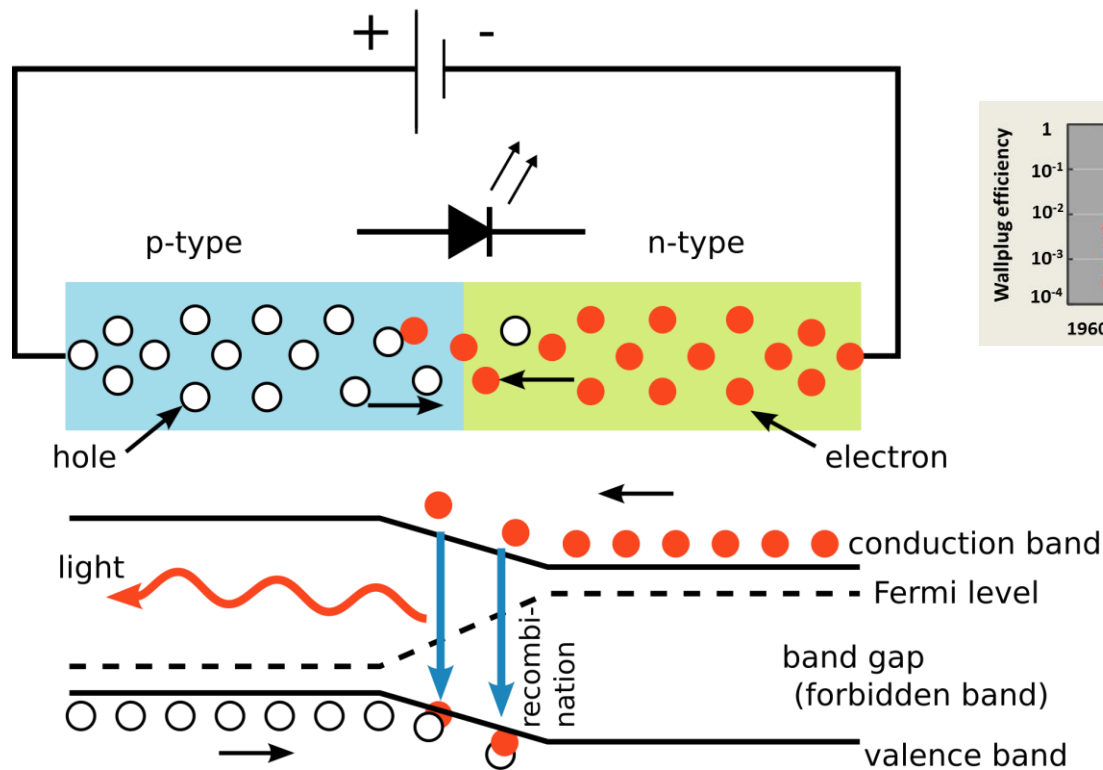


p-n Junction as Rectifier



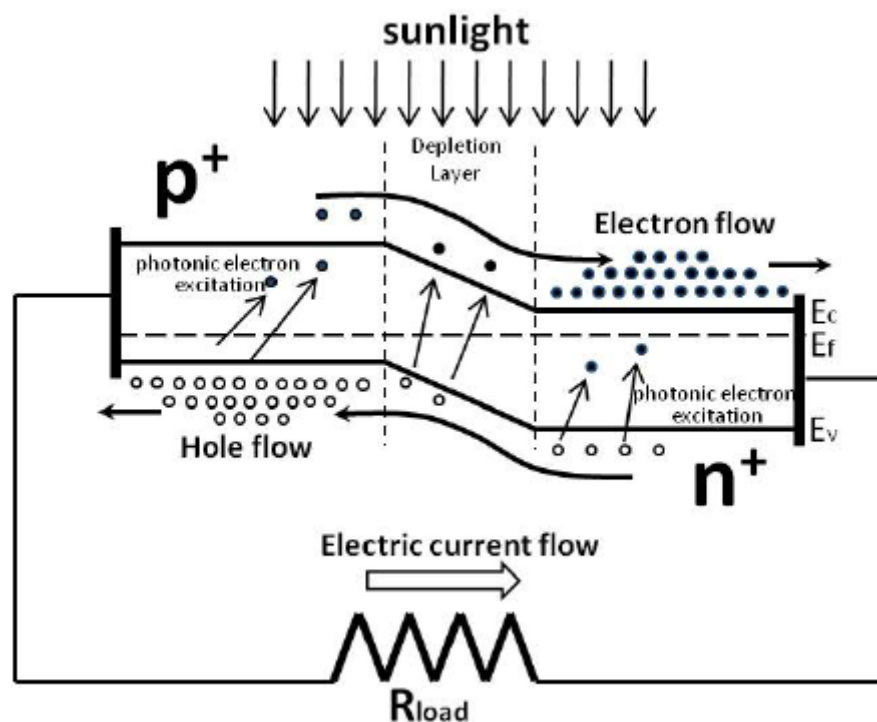
(Image: <https://www.fiberoptics4sale.com/blogs/wave-optics/semiconductor-junctions>)

Light-Emitting Diodes (LEDs)



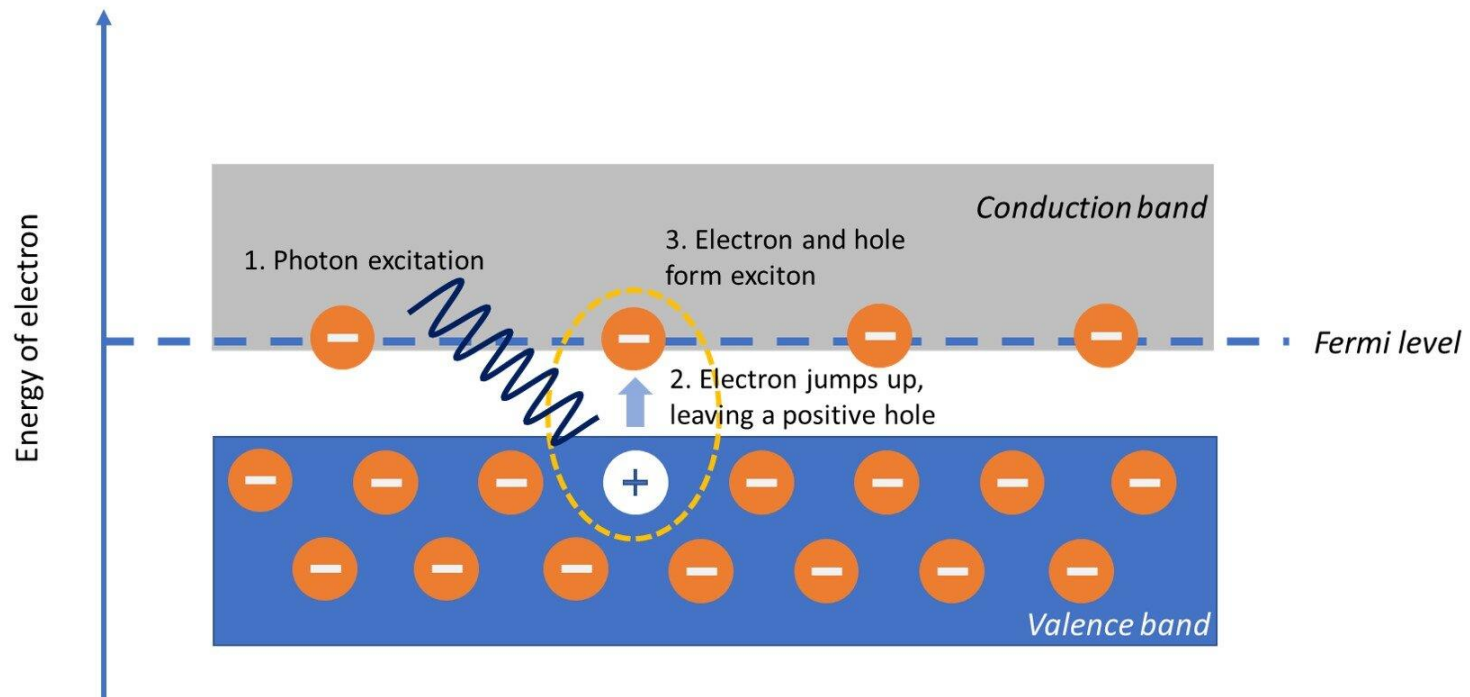
Nobel prize in
Physics 2014

Solar Cells



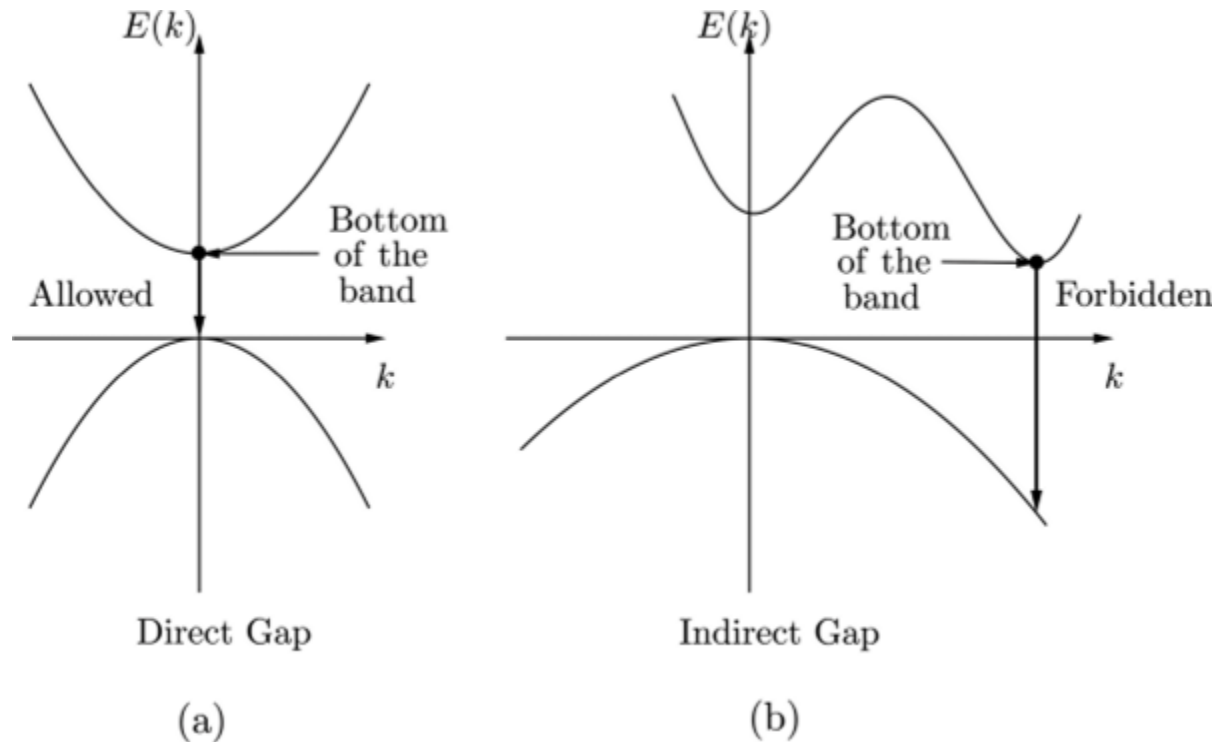
Exciton

(electron-hole coupling)



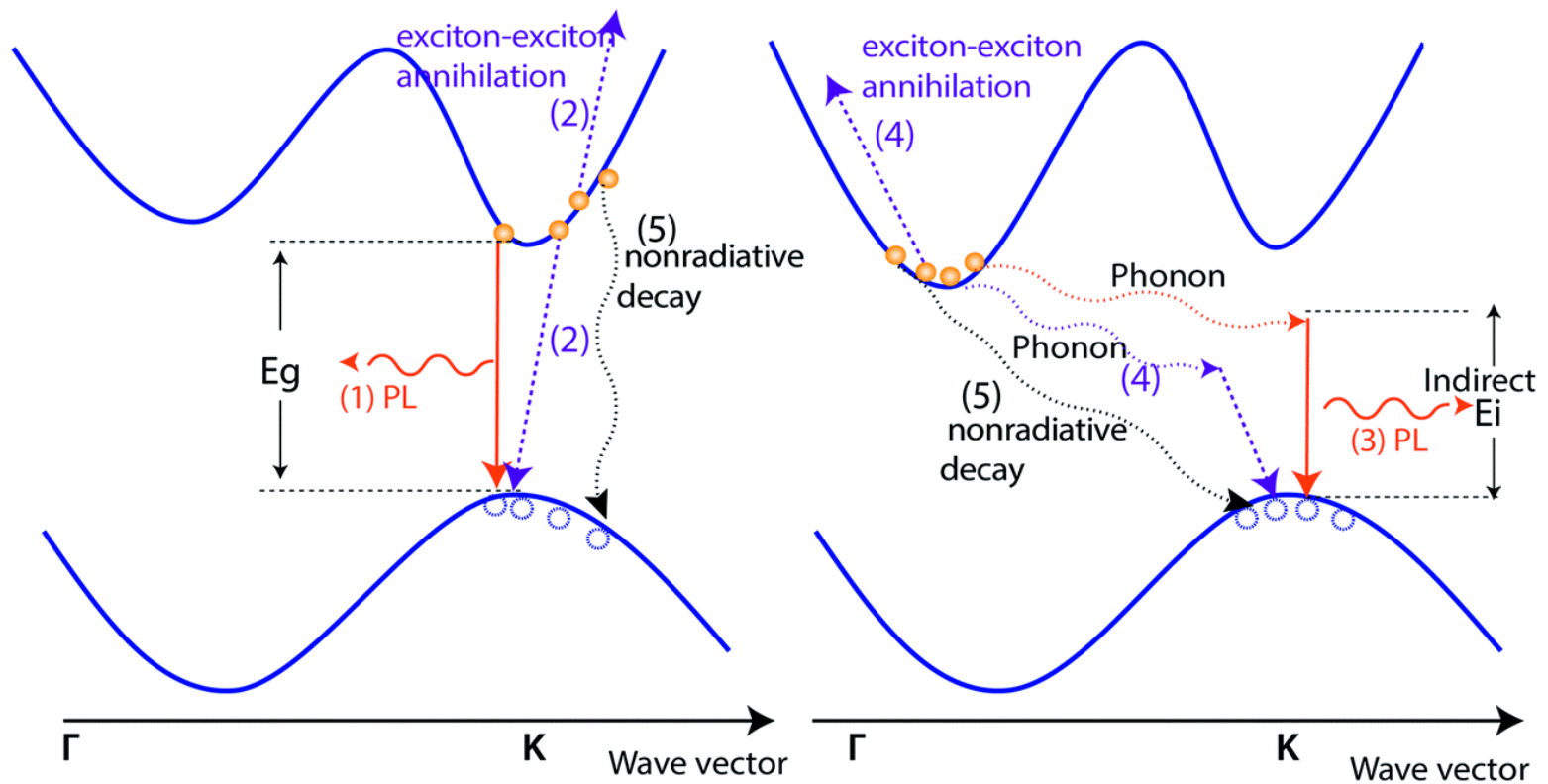
(Image: <https://phys.org/news/2022-03-fast-moving-excitons-metal-potential-digital.html>)

Direct and Indirect Band Gaps



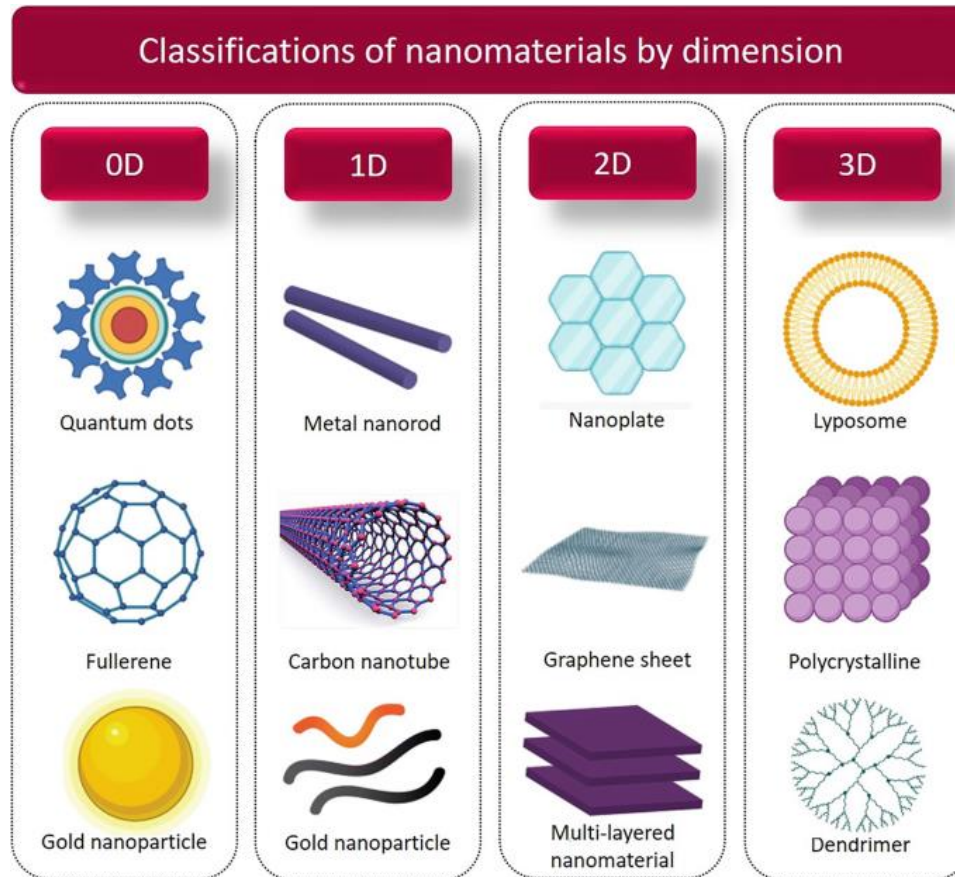
(Image: https://ebrary.net/158530/mathematics/direct_indirect_band_optical_transitions)

Exciton Dynamics



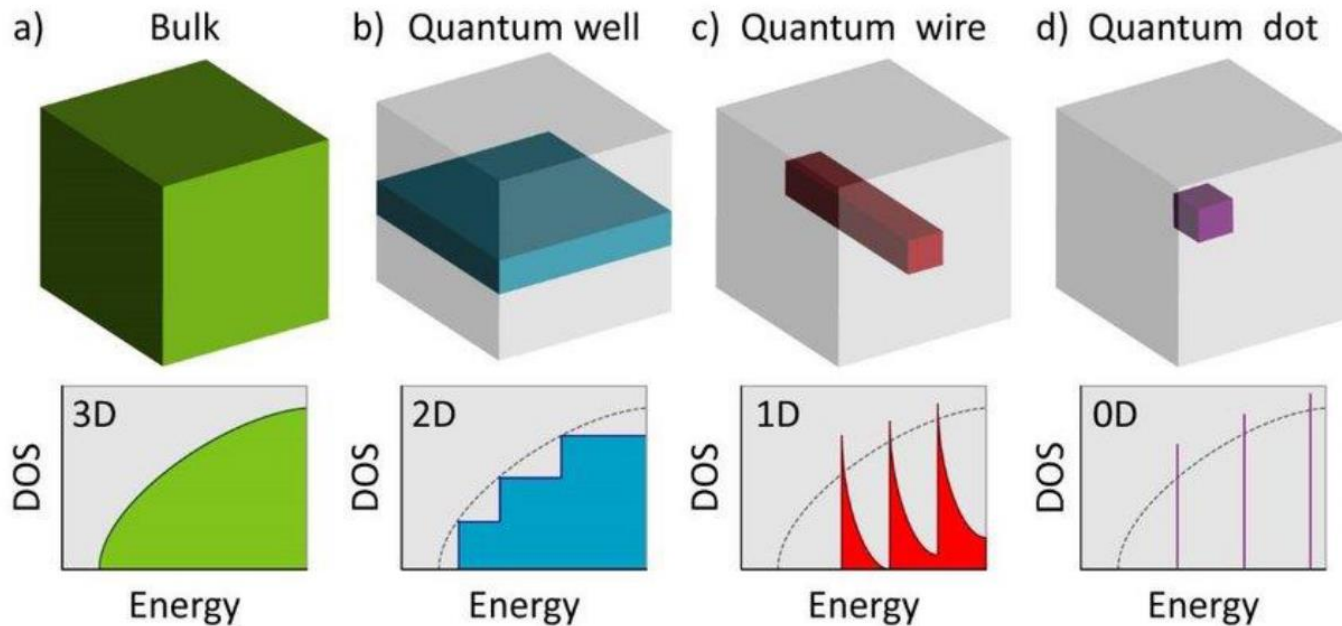
(Image: Yuan and Huang, *Nanoscale* (2015), Fig. 4)

Nanomaterials



(Image: Zainon et al., *Toxicity of Nanomaterials* (2022))

Quantum Confinement on DOS



DOS of Free Particles

Consider a classical particle in d dimensions. The number of states is

$$N_{\text{states}} = \frac{1}{h^d} \int d^d \vec{r} d^d \vec{p} = \frac{V}{h^d} \int d^d \vec{p}$$

DOS of 3D Free Particles

$$N_{\text{states}} = \frac{1}{h^d} \int d^d \vec{r} d^d \vec{p} = \frac{V}{h^d} \int d^d \vec{p}$$

For $d = 3$,

$$\int d^3 \vec{p} = \iiint p^2 \sin \theta dp d\theta d\phi = \int 4\pi p^2 dp$$

As $E = p^2/2m$, $p^2 dp = m\sqrt{2mE} dE$ and

$$N_{\text{states}} = \frac{V}{h^3} \int 4\pi m\sqrt{2mE} dE$$

DOS of 3D Free Particles

$$N_{\text{states}} = \frac{V}{h^3} \int 4\pi m \sqrt{2mE} \, dE$$

On the other hand,

$$N_{\text{states}} = \int g(E) \, dE$$

Hence,

$$g(E) = \frac{2\pi V}{h^3} (2m)^{3/2} E^{1/2}$$

DOS of 2D Free Particles

$$N_{\text{states}} = \frac{1}{h^d} \int d^d \vec{r} d^d \vec{p} = \frac{V}{h^d} \int d^d \vec{p}$$

For $d = 2$,

$$\int d^2 \vec{p} = \iint p \, dp d\theta = \int 2\pi p \, dp$$

As $E = p^2/2m$, $p \, dp = m \, dE$ and

$$N_{\text{states}} = \frac{A}{h^2} \int 2\pi m \, dE$$

DOS of 2D Free Particles

$$N_{\text{states}} = \frac{A}{h^2} \int 2\pi m \, dE$$

On the other hand,

$$N_{\text{states}} = \int g(E) \, dE$$

Hence,

$$g(E) = \frac{2\pi A}{h^2} m \text{ (constant)}$$

DOS of 1D Free Particles

$$N_{\text{states}} = \frac{1}{h^d} \int d^d \vec{r} d^d \vec{p} = \frac{V}{h^d} \int d^d \vec{p}$$

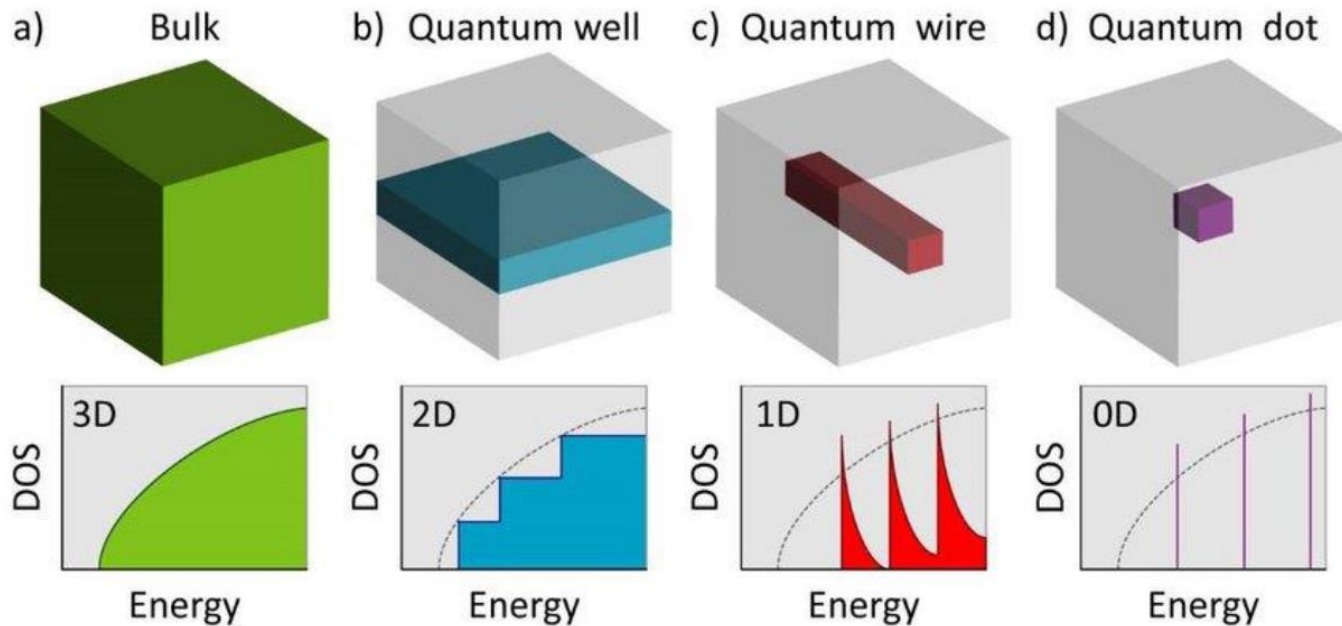
For $d = 1$, as $E = p^2/2m$,

$$dp = \frac{m}{\sqrt{2mE}} dE$$

$$N_{\text{states}} = \frac{L}{h} \int \frac{m}{\sqrt{2mE}} dE$$

$$g(E) = \frac{Lm}{h\sqrt{2mE}}$$

Quantum Confinement on DOS

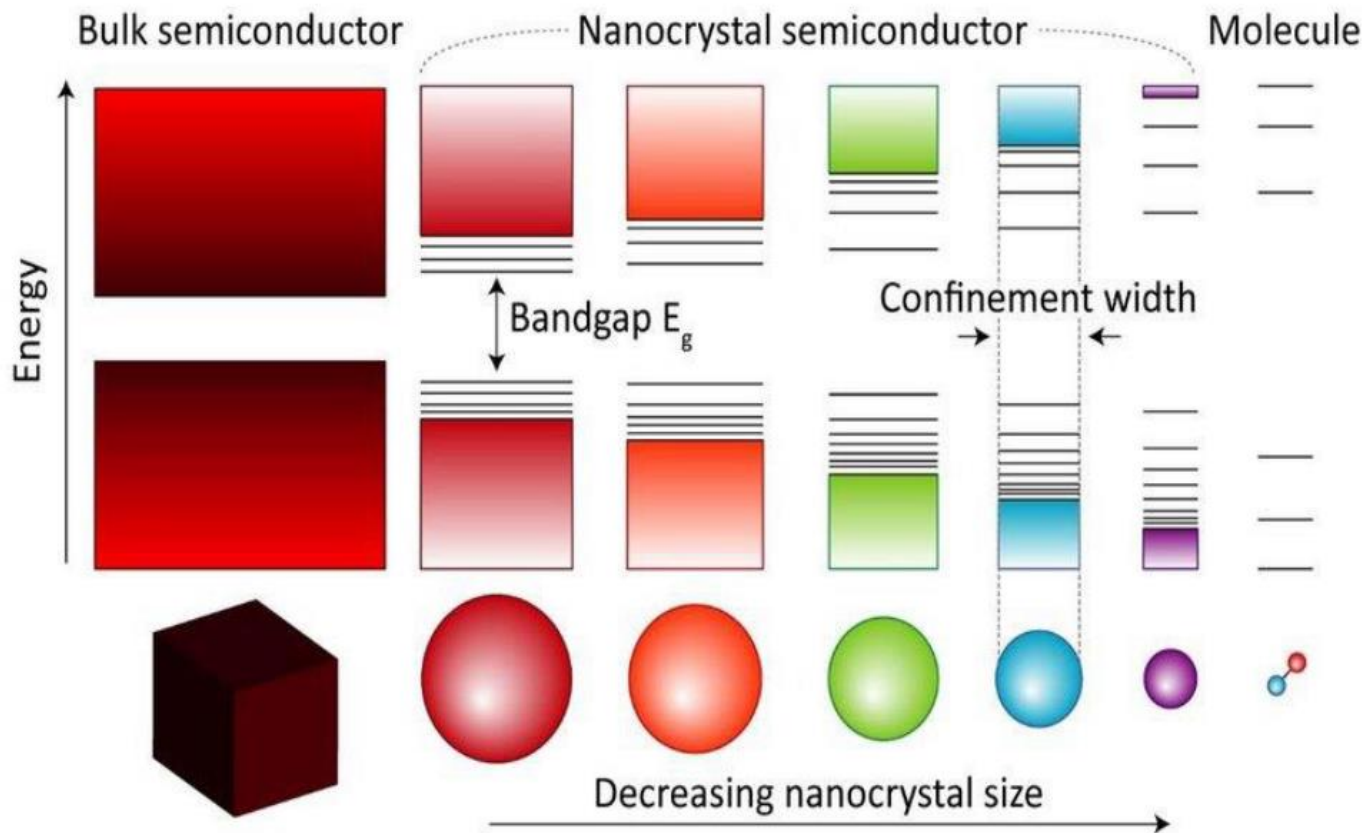


$$g(E) \propto E^{1/2}$$

$$g(E) \propto E^0$$

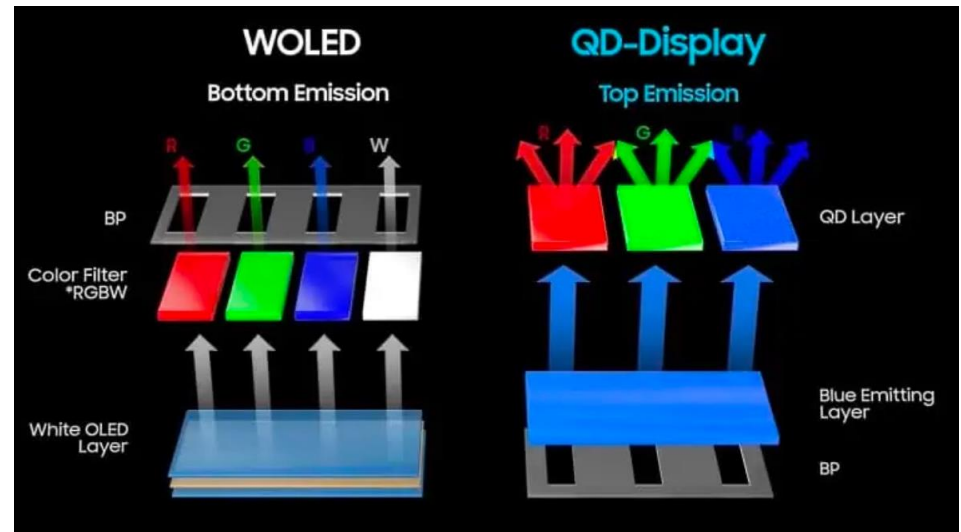
$$g(E) \propto E^{-1/2}$$

Quantum Dots: Size Matters



(Image: <https://www.nanoinnovation2020.eu/home/PPT/Synthesis%20characteristics%20uses%20and%20detection%20of%20engineered%20nanostructured%20materials.pdf>)

Application of Quantum Dots



Nobel prize in Chemistry 2023

(Images: <https://avantama.com/what-is-a-quantum-dot/>
<https://www.sammobile.com/news/qd-oled-tv-features-explained/>)